

Engineering Mechanics: ME101

Lecture 6: Statics



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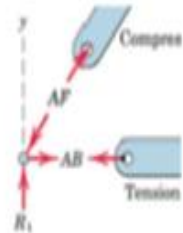
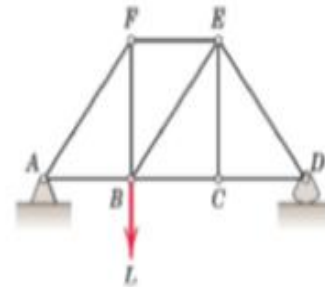
Contents

- ❖ Introduction to method of sections
- ❖ Plane truss determinacy and analysis methods
- ❖ Frames and Machines
- ❖ Analysis of a Frame

Structural Analysis of Plane Truss

❑ There are two methods of analysis of trusses namely:- (1) Method of joints and (2) Method of sections.

❖ **Method of Joints:** only two of three equilibrium equations were applied at each joint because the procedures involve concurrent forces at each joint



→ Calculations from joint to joint

→ More time and effort required

❖ **Method of Sections**

Take advantage of the 3rd or moment equation of equilibrium by selecting an entire section of truss

→ Equilibrium under non-concurrent force system

→ Not more than 3 members whose forces are unknown should be cut in a single section since we have only 3 independent equilibrium equations

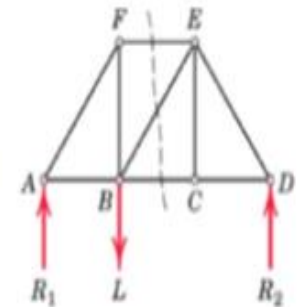


Figure 2

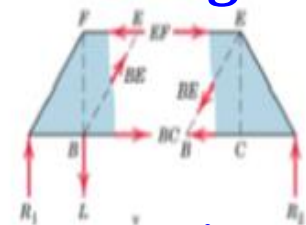


Figure 3

Principal of Method of Sections

Principal : If a body is in equilibrium, then any part of the body is also in equilibrium.

- Forces in few particular member can be directly found out quickly without solving each joint of the truss sequentially
- Method of Sections and Method of Joints can be conveniently combined
- A section need not be straight.
- More than one section can be used to solve a given problem

❖ **Note:** Cut the truss in such a way that section should not pass through more than three members.



Trusses Made of Several Simple Trusses

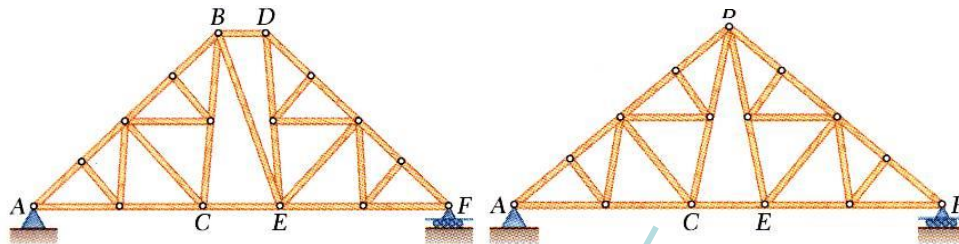


Figure 1

Figure 2

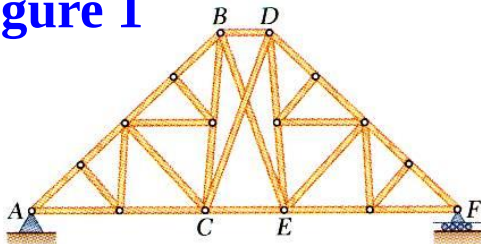
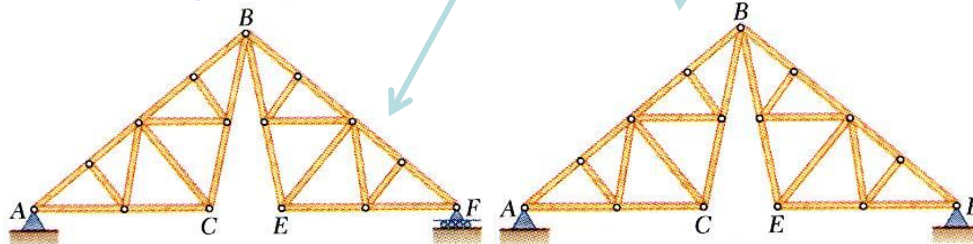


Figure 3



non-rigid

$$m < 2n - 3$$

Figure 4

rigid

$$m < 2n - 4$$

Figure 5

1. Compound trusses are statically determinate, rigid, and completely constrained.

$$m = 2n - 3$$

2. Truss contains a *redundant member* and is *statically indeterminate*.

$$m > 2n - 3$$

3. Additional reaction forces may be necessary for a rigid truss.

4. Necessary but insufficient condition for a compound truss to be statically determinate, rigid, and completely constrained,

$$m + r = 2n$$

How to approach problems on Method of Sections



Method of Sections

Find out the reactions from equilibrium of whole truss



To find force in member BE

Cut an imaginary section (dotted line)

Each side of the truss section should remain in equilibrium



For calculating force on member EF, take moment about B

Take moment about E, for calculating force BC



Now apply $\sum F_y = 0$ to obtain forces on the members BE

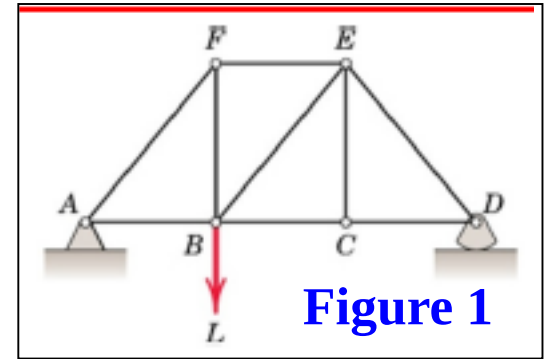


Figure 1

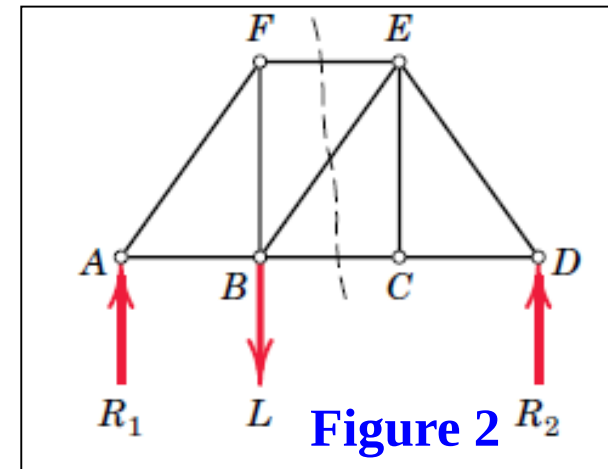


Figure 2

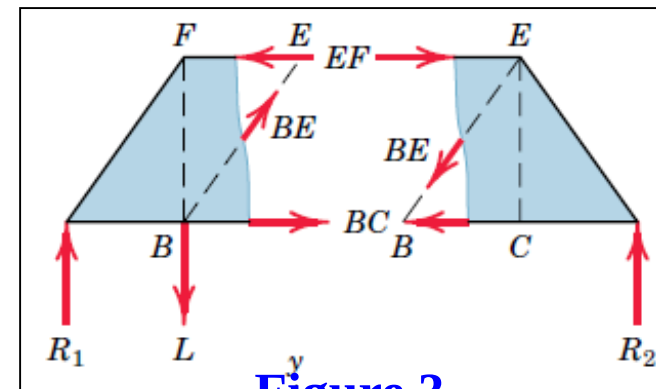


Figure 3

Method of sections (*Problem*)

Figure 1

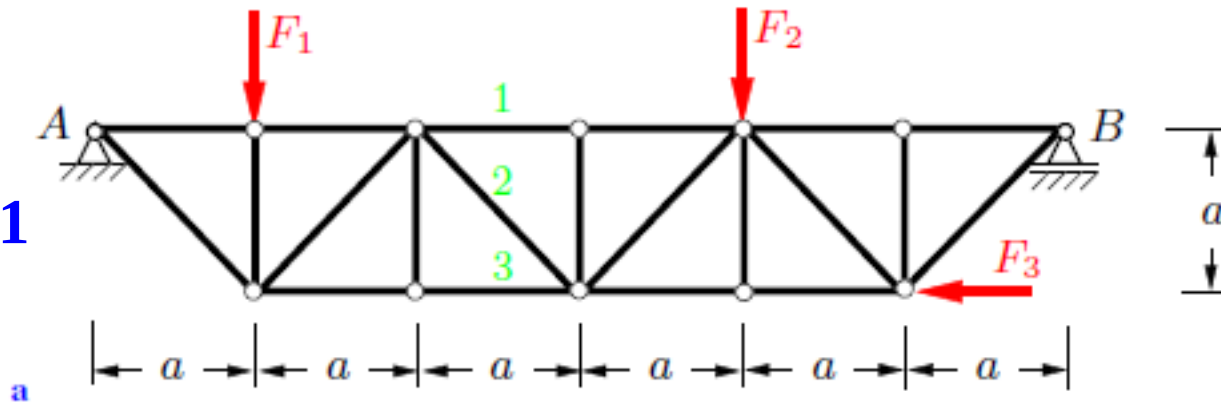
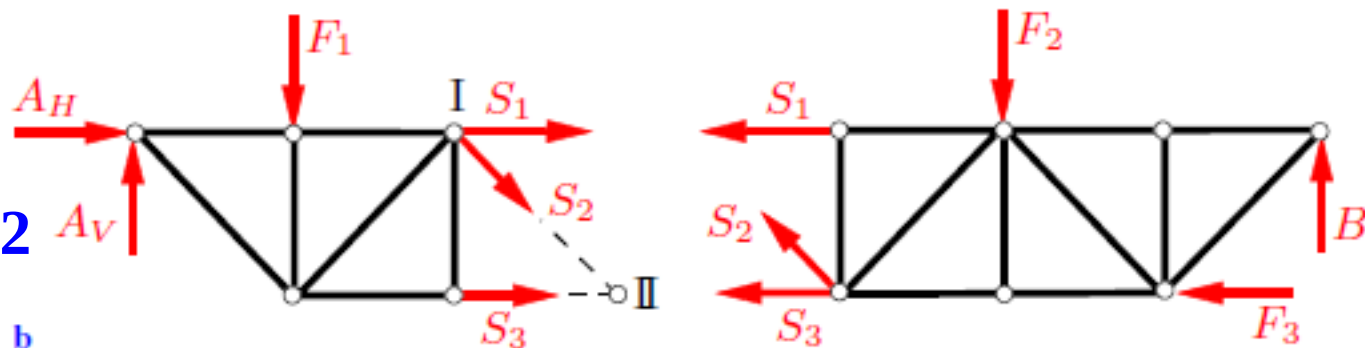


Figure 2



$$\text{I} : -2a A_V + a F_1 + a S_3 = 0 \quad \rightarrow \quad S_3 = 2A_V - F_1, \quad (\text{i})$$

$$\text{II} : -3a A_V - a A_H + 2a F_1 - a S_1 = 0 \quad (\text{ii})$$

$$\rightarrow \quad S_1 = 2F_1 - 3A_V - A_H, \quad (\text{iii})$$

$$\uparrow : A_V - F_1 - \frac{1}{2}\sqrt{2} S_2 = 0 \quad \rightarrow \quad S_2 = \sqrt{2}(A_V - F_1). \quad (\text{iv})$$

Sample Problem

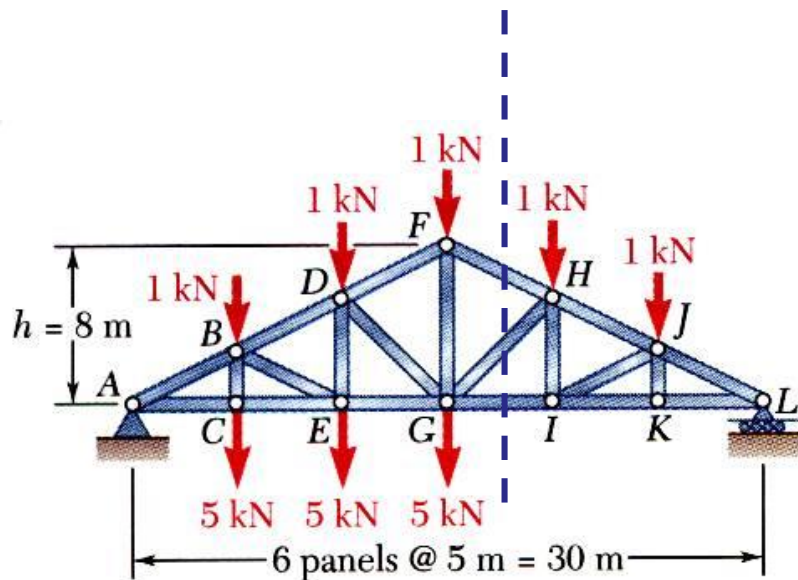


Figure 1

Determine the force in members FH , GH , and GI .

SOLUTION:

- List the steps for solving this problem.

1. Draw the FBD for the entire truss. Apply the equilibrium conditions and solve for the reactions at A and L.
2. Make a cut through members FH , GH , and GI and take the right-hand section as a free body (the left side would also be good).
3. Apply the conditions for static equilibrium to determine the desired member forces.

Sample Problem

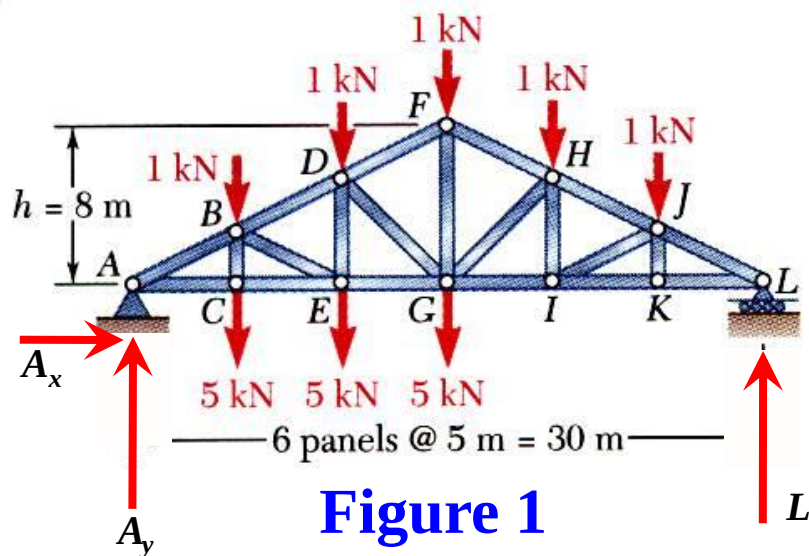


Figure 1

SOLUTION:

- Take the entire truss as a free body. Apply the conditions for static equilibrium to solve for the reactions at A and L.

$$\sum M_A = 0 = -(5 \text{ m})(6 \text{ kN}) - (10 \text{ m})(6 \text{ kN}) - (15 \text{ m})(6 \text{ kN}) - (20 \text{ m})(1 \text{ kN}) - (25 \text{ m})(1 \text{ kN}) + (25 \text{ m})L \quad (\text{i})$$

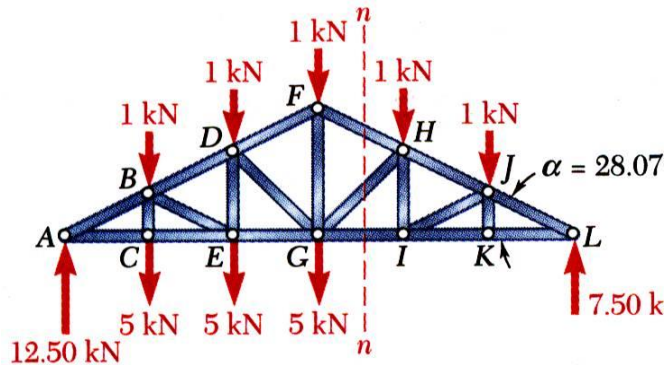
$$L = 7.5 \text{ kN}$$

$$\sum F_y = 0 = -20 \text{ kN} + L + A_y \quad (\text{ii})$$

$$A_y = 12.5 \text{ kN}$$

$$\sum F_x = 0 = A_x \quad (\text{iii})$$

Sample Problem



- Make a cut through members FH , GH , and GI and take the right-hand section as a free body.
Draw this FBD.

- What is the one equilibrium equation that could be solved to find F_{GI} ?

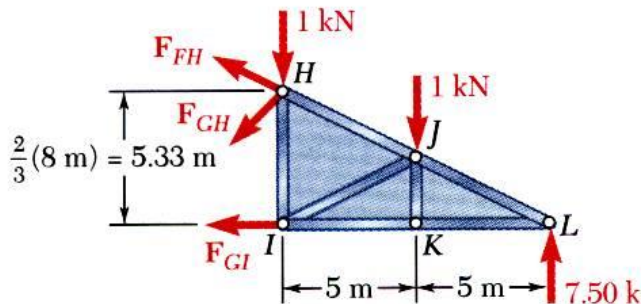
- Sum of the moments about point H :

$$\sum M_H = 0 \quad \text{(iv)}$$

$$(7.50 \text{ kN})(10 \text{ m}) - (1 \text{ kN})(5 \text{ m}) - F_{GI}(5.33 \text{ m}) = 0$$

$$F_{GI} = +13.13 \text{ kN}$$

$$F_{GI} = 13.13 \text{ kN } T$$



Sample Problem

- F_{FH} is shown as its components. What one equilibrium equation will determine F_{FH} ?

$$\tan \alpha = \frac{FG}{GL} = \frac{8 \text{ m}}{15 \text{ m}} = 0.5333 \quad (\text{v}) \quad \alpha = 28.07^\circ$$

$$\sum M_G = 0 \quad (\text{vi})$$

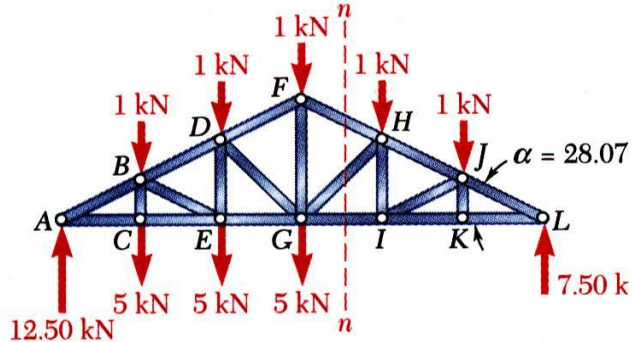
$$(7.5 \text{ kN})(15 \text{ m}) - (1 \text{ kN})(10 \text{ m}) - (1 \text{ kN})(5 \text{ m}) + (F_{FH} \cos \alpha)(8 \text{ m}) = 0$$

$$F_{FH} = -13.81 \text{ kN}$$

$$F_{FH} = 13.81 \text{ kN } C$$

Figure 2

Sample Problem



- There are many options for finding F_{GH} at this point (e.g., $\Sigma F_x = 0$, $\Sigma F_y = 0$). Here is one more:

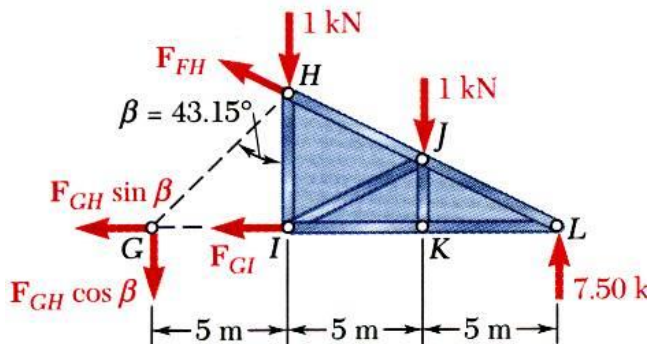


Figure 3

$$\tan \beta = \frac{GI}{HI} = \frac{5 \text{ m}}{\frac{2}{3}(8 \text{ m})} = 0.9375 \quad \beta = 43.15^\circ$$

$$\Sigma M_L = 0 \quad \text{(vii)}$$

$$(1 \text{ kN})(10 \text{ m}) + (1 \text{ kN})(5 \text{ m}) + (F_{GH} \cos \beta)(15 \text{ m}) = 0$$

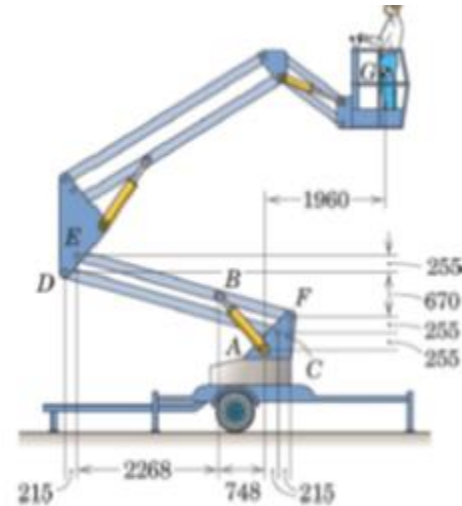
$$F_{GH} = -1.371 \text{ kN}$$

$$F_{GH} = 1.371 \text{ kN C}$$

Frames and Machines

❖ A structure is called a Frame or Machine if at least one of its individual members is a multi-force member

- member with 3 or more forces acting, or
- member with 2 or more forces and 1 or more couple acting



- ❖ **Frames:** generally stationary and are used to support loads
- ❖ **Machines:** contain moving parts and are designed to transmit and alter the effect of forces acting
- ❖ **Multi-force members:** the forces in these members in general will not be along the directions of the members
→ methods used in simple truss analysis cannot be used

Forces of Interactions

- ❑ Free Body Diagrams along with the forces acting on it are represented as:
 - ✓ force components must be consistently represented in opposite directions on the separate FBDs (Ex: Pin at A).
 - ✓ apply action-and-reaction principle (Ex: Ball & Socket at A).
 - ✓ **Vector notation:** use plus sign for an action and a minus sign for the corresponding reaction.

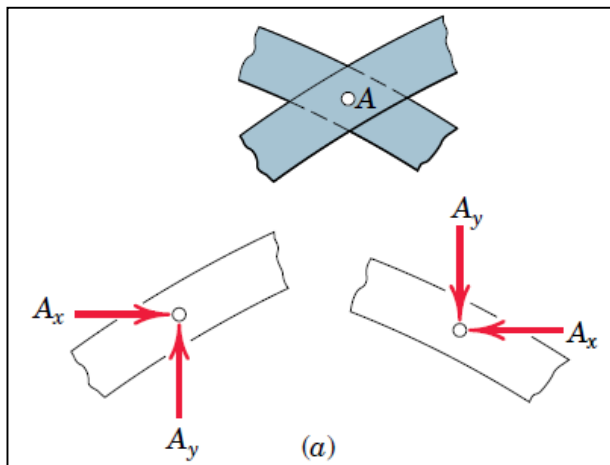


Figure 1

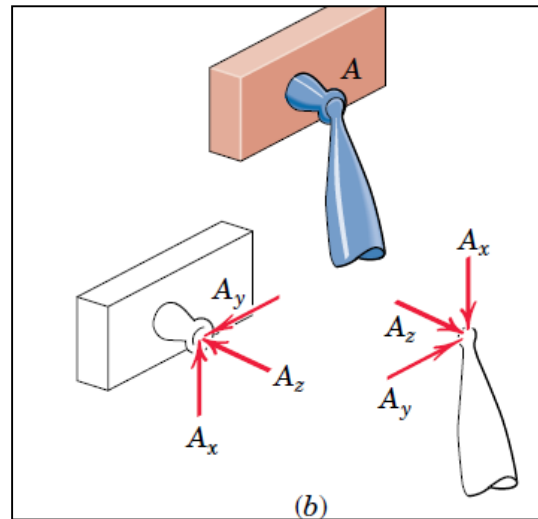


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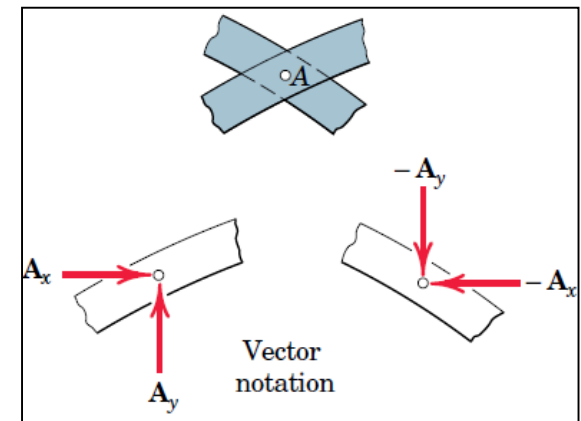


Figure 3

Analysis of a Frame

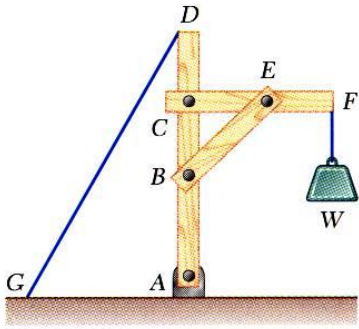


Figure 1

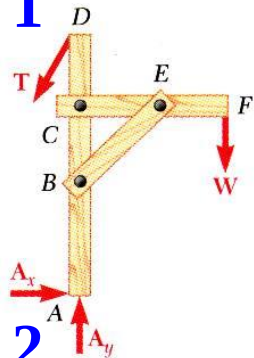


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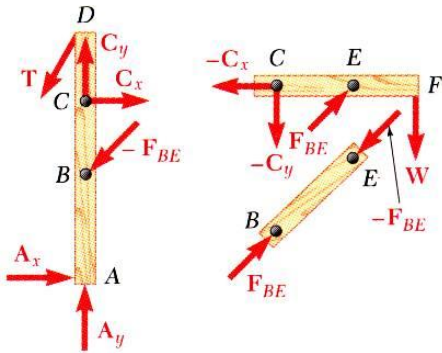
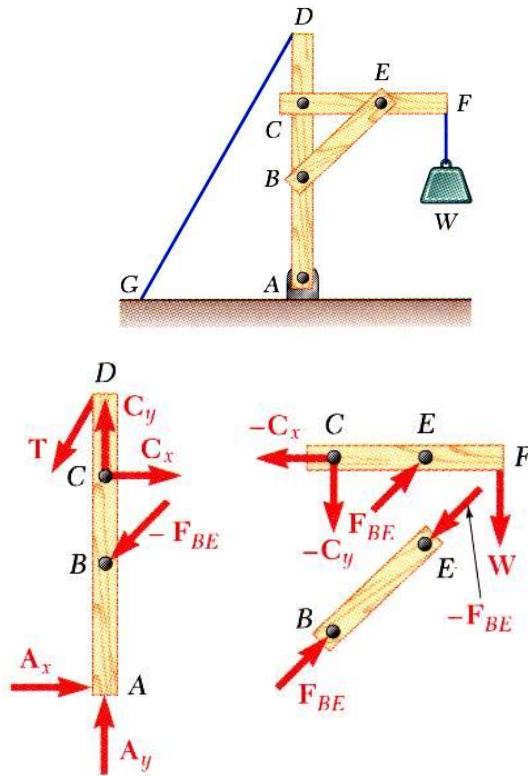


Figure 3

1. *Frames and machines* are structures with at least one *multiforce* (>2 forces) member. **Frames** are designed to support loads and are usually stationary. **Machines** contain moving parts and transmit and modify forces.
2. A **free body diagram** of the complete frame is used to determine the **external forces** acting on the frame.
3. **Internal forces are determined** by dismembering the frame and creating free-body diagrams for each component.
4. Forces on **two force members** have known lines of **action** but unknown magnitude and sense.

Analysis of a Frame (*contd.*)



4. Forces on two force members have known lines of action but unknown magnitude and sense.
5. Forces on multiforce members have unknown magnitude and line of action. They must be represented with two unknown components.
6. Forces between connected components are equal, have the same line of action, and opposite sense.

Figure 3

Frames Which Cease To Be Rigid When Detached From Their Supports

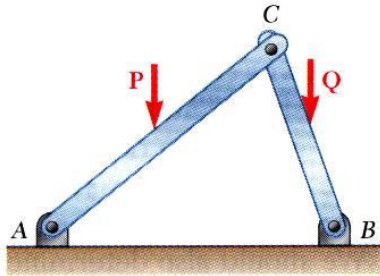


Figure 1

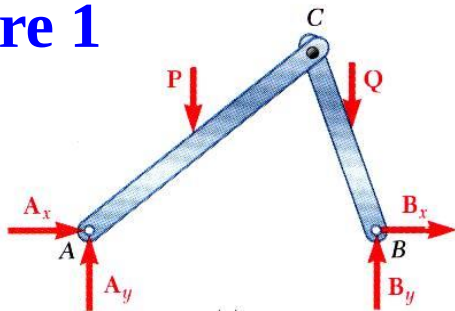


Figure 2

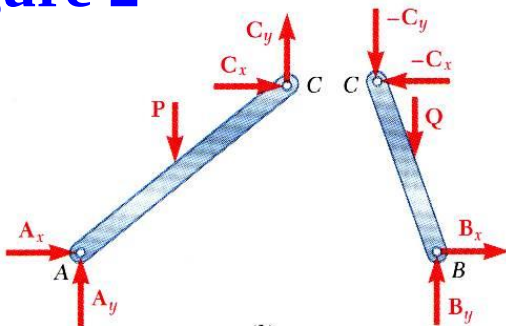
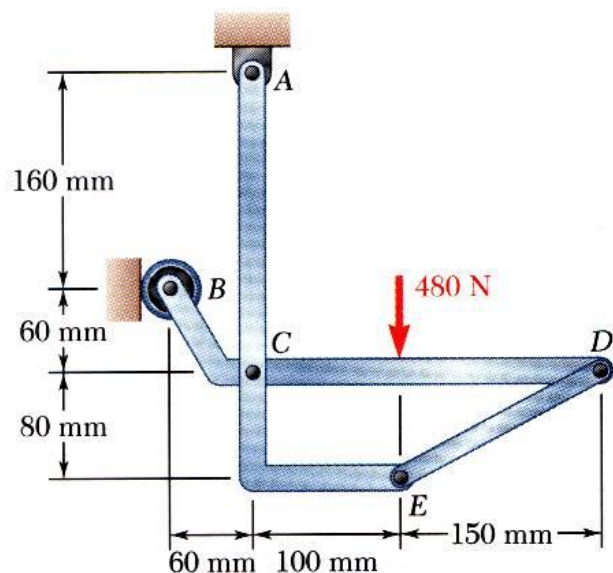


Figure 3

1. Some frames may collapse if removed from their supports. Such frames cannot be treated as rigid bodies.
2. A free-body diagram of the complete frame indicates four unknown force components which cannot be determined from the three equilibrium conditions (statically indeterminate).
3. The frame must be considered as two distinct, but related, rigid bodies.
4. With equal and opposite reactions at the contact point between members, the two free-body diagrams show 6 unknown force components.
5. Equilibrium requirements for the two rigid bodies yield 6 independent equations. Thus, taking the frame apart made the problem solvable.

Sample Problem



SOLUTION:

1. Create a **free body diagram** for the complete frame and solve for the **support reactions**.

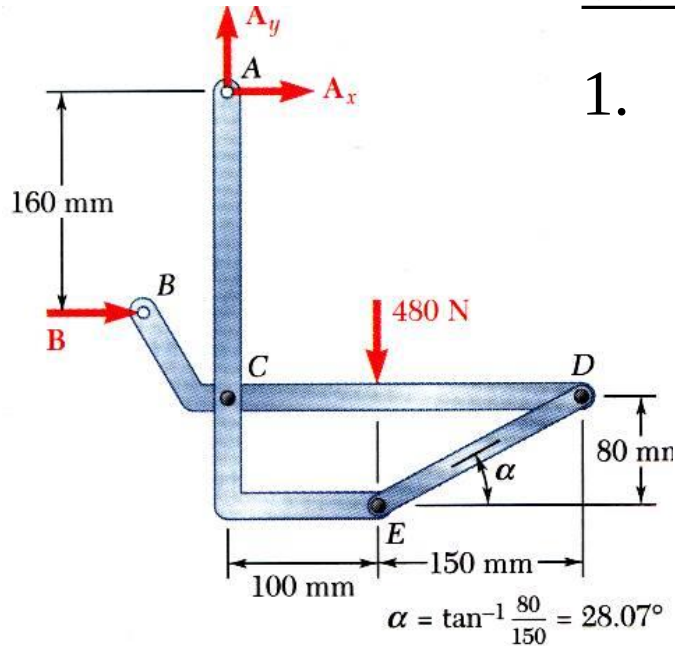
□ Members ACE and BCD are connected **by a pin at C** and by the link DE . For the loading shown,

- ✓ determine the force in link DE and the components of the force exerted at C on member BCD .

Sample Problem

SOLUTION:

1. Create a free-body diagram for the complete frame and solve for the support reactions.



$$\sum F_y = 0 = A_y - 480 \text{ N}$$

$$A_y = 480 \text{ N}$$

$$\sum M_A = 0 = -(480 \text{ N})(100 \text{ mm}) + B(160 \text{ mm})$$

$$B = 300 \text{ N}$$

$$\sum F_x = 0 = B + A_x$$

$$A_x = -300 \text{ N}$$

$$A_x = -300 \text{ N}$$

Sample Problem

SOLUTION (cont.):

2. Create a free body diagram for member *BCD* (since the problem asked for forces on this body). **Choose the best FBD and Justify your choice.**

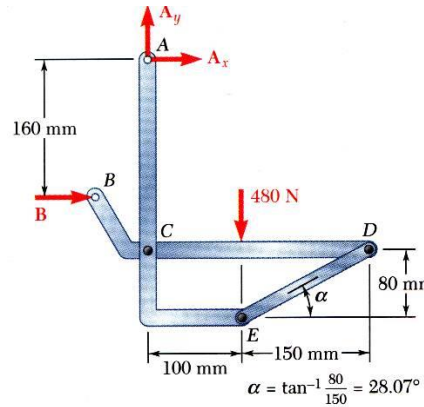


Figure 1

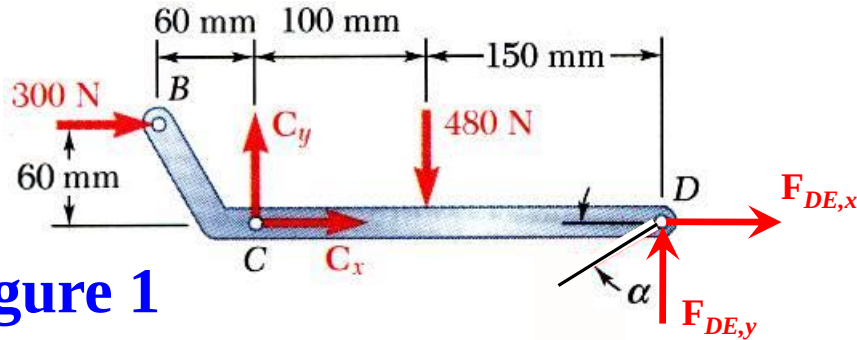


Figure 2

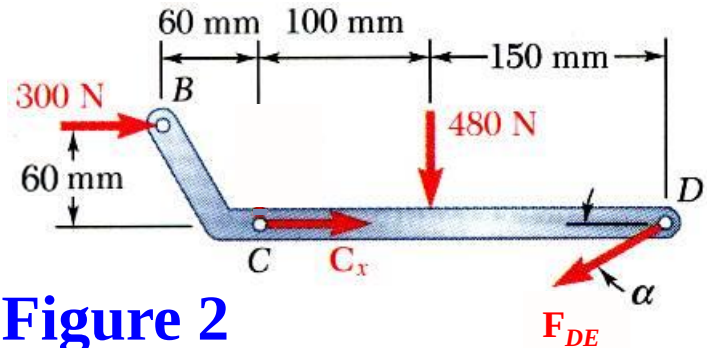


Figure 3

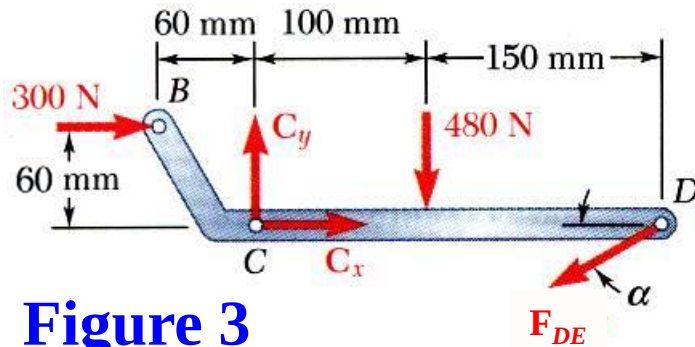
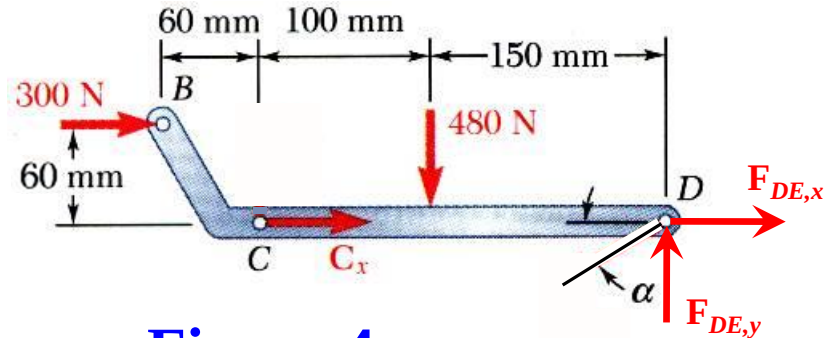


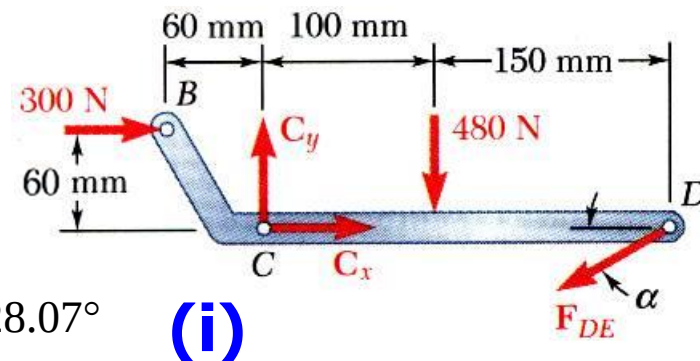
Figure 4



Sample Problem

SOLUTION (cont.):

3. Using the best FBD for member *BCD*, what is the one equilibrium equation that can directly find F_{DE} ? Please discuss.



$$\alpha = \tan^{-1} \frac{80}{150} = 28.07^\circ \quad \text{(i)}$$

$$\sum M_C = 0 = (F_{DE} \sin \alpha)(250 \text{ mm}) + (300 \text{ N})(60 \text{ mm}) + (480 \text{ N})(100 \text{ mm}) \quad \text{(ii)}$$

$$F_{DE} = -561 \text{ N}$$

$$F_{DE} = 561 \text{ N } C$$

- Sum of forces in the x and y directions may be used to find the force components at C .

$$\sum F_x = 0 = C_x - F_{DE} \cos \alpha + 300 \text{ N} \quad \text{(iii)}$$

$$0 = C_x - (-561 \text{ N}) \cos \alpha + 300 \text{ N}$$

$$C_x = -795 \text{ N}$$

$$\sum F_y = 0 = C_y - F_{DE} \sin \alpha - 480 \text{ N} \quad \text{(iv)}$$

$$0 = C_y - (-561 \text{ N}) \sin \alpha - 480 \text{ N}$$

$$C_y = 216 \text{ N}$$

Sample Problem

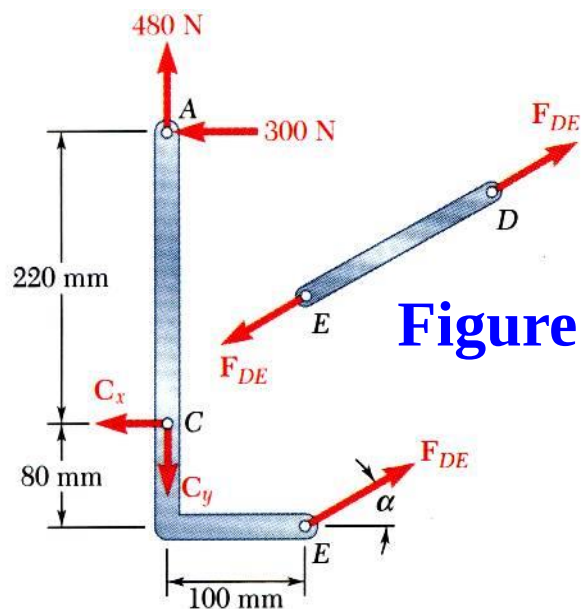


Figure 5

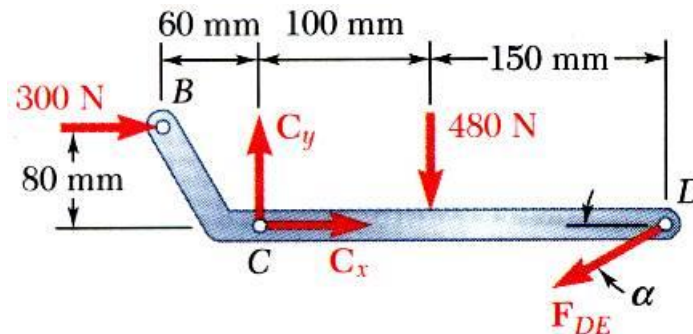
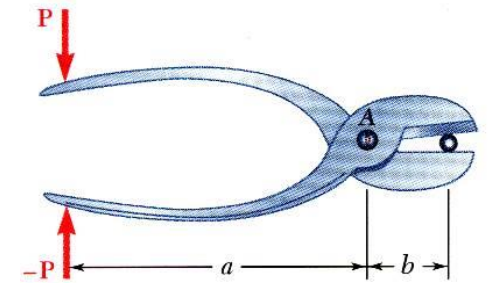


Figure 6

- With member *ACE* as a free body with no additional unknown forces, check the solution by summing moments about *A*.

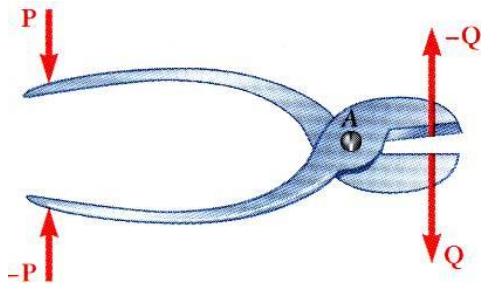
$$\begin{aligned}\sum M_A &= (F_{DE} \cos \alpha)(300 \text{ mm}) + (F_{DE} \sin \alpha)(100 \text{ mm}) - C_x(220 \text{ mm}) \quad \text{(v)} \\ &= (-561 \cos \alpha)(300 \text{ mm}) + (-561 \sin \alpha)(100 \text{ mm}) - (-795)(220 \text{ mm}) = 0 \quad \text{(vi)} \\ &\quad \text{(checks)}\end{aligned}$$

Machines

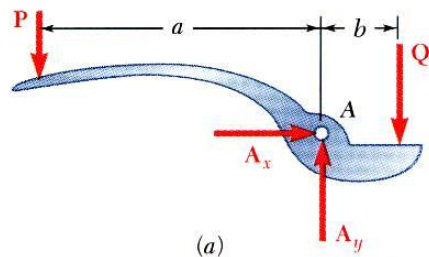


1. Machines are structures designed to **transmit and modify forces**. Typically they transform *input forces ($\mathbf{P}$)* into *output forces ($\mathbf{Q}$)*.

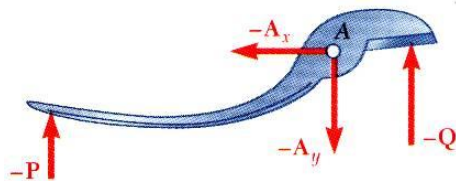
- Given the magnitude of $\mathbf{P}$, determine the magnitude of $\mathbf{Q}$.



2. Create a free-body diagram of the complete machine, including the reaction that the **wire exerts**.



3. The machine is **a nonrigid structure**. Use one of the components as a free-body. **Discuss why the forces at A are such.**

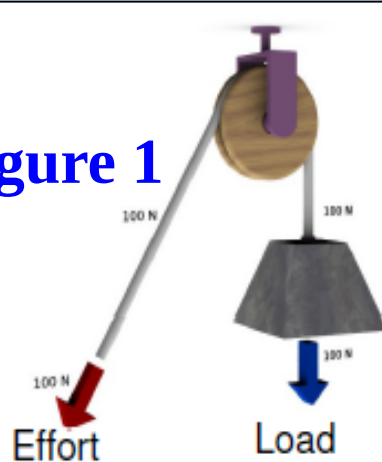


- Sum moments about A ,

$$\sum M_A = 0 = aP - bQ \quad Q = \frac{a}{b}P \quad \text{(i)}$$

Effort, Load, Velocity Ratio & Mechanical Advantage

Figure 1



Fixed Pulley

Effort = Load
 → Mechanical Advantage = 1

Distance moved by effort is equal to the distance moved by the load.

→ Velocity Ratio = 1

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Figure 2



Movable Pulley

Effort = Load/2
 → Mechanical Advantage = 2

Distance moved by effort is twice the distance moved by the load (both rope should also accommodate the same displacement by which the load is moved).

→ Velocity Ratio = 2

Compound Pulley

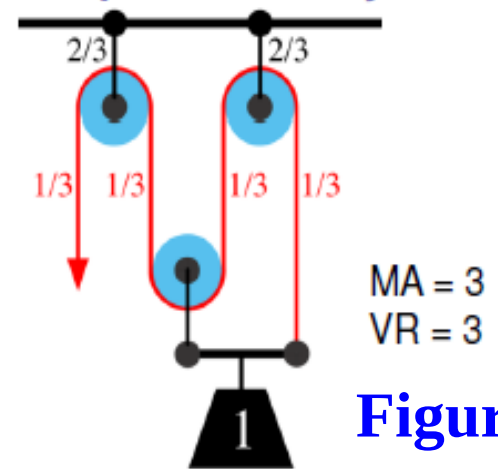


Figure 3

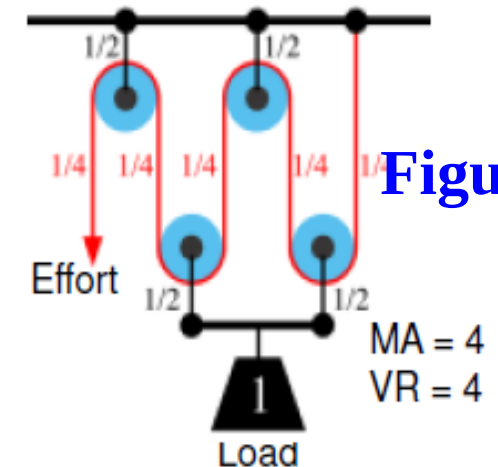
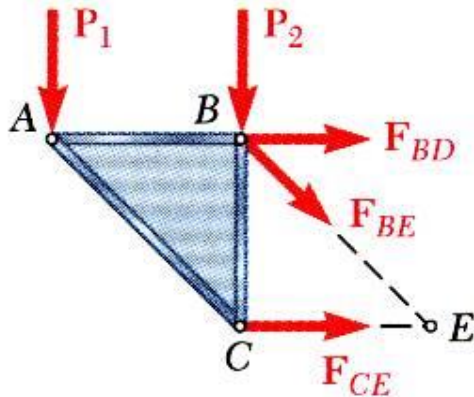
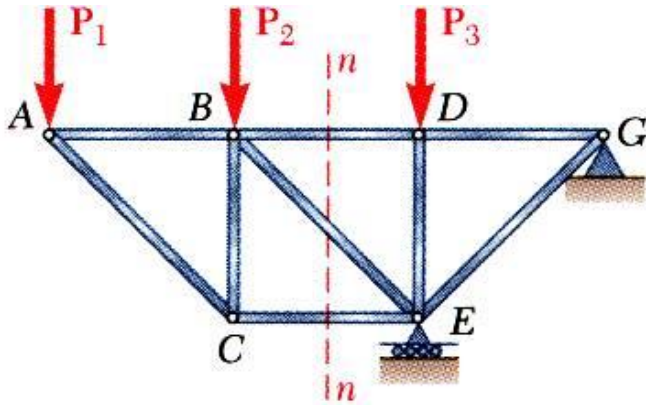


Figure 4

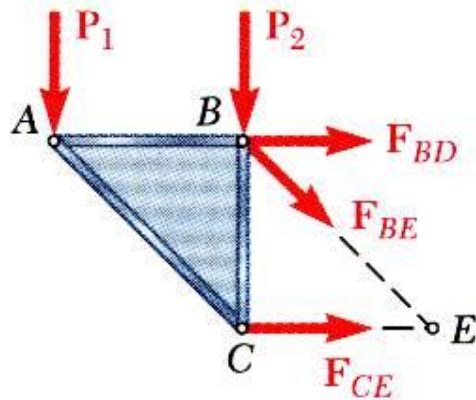
❖ **Mechanical Advantage (MA) = Load/Effort**

Analysis of Trusses by the Method of Sections

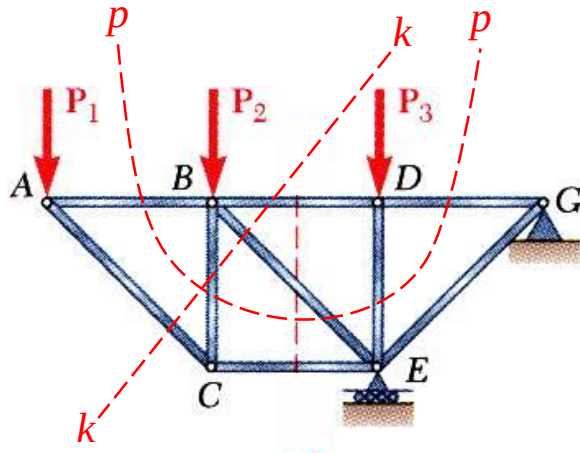


- When the force in only one member or the forces in a very few members are desired, the *method of sections* works well.
- To determine the force in member BD , form a *section* by “cutting” the truss at n - n and create a free body diagram for the left side.
- A FBD could have been created for the right side, but **why is this a less desirable choice?**
- Notice that the exposed internal forces are all *assumed* to be in tension.
- With only three members cut by the section, the equations for static equilibrium may be applied to determine the unknown member forces, including F_{BD} .

Analysis of Trusses by the Method of Sections



- Using the left-side FBD, write one equilibrium equation that can be solved to find F_{BD} .



- Assume that the initial section cut was made using line $k-k$. Why would this be a poor choice?
- Notice that *any* cut may be chosen, so long as the cut creates a separated section.
- So, for example, this cut with line $p-p$ is acceptable.

END